SCHOOL OF ENGINEERING AND TECHNOLOGY

COURSEWORK FOR

BSC (HONS) IN COMPUTER SCIENCE
BSC (HONS) INFORMATION TECHNOLOGY
BACHELOR OF SOFTWARE ENGINEERING (HONS)
BSC (HONS) INFORMATION SYSTEMS (DATA ANALYTICS)
BSC (HONS) INFORMATION TECHNOLOGY (COMPUTER NETWORKING AND SECURITY)
YEAR 1; ACADEMIC SESSION APRIL 2025

SEG1201: DATABASE FUNDAMENTALS

Answer Template and SQL Script and Video Presentation Submission: Due date :Week 14 TUESDAY (5 AUGUST 2025), 11.59 pm

SEG1201: DATABASE FUNDAMENTALS

GROUP NAME: 12_3

MEMBER:

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YOUTUBE LINK:

https://www.youtube.com/watch?v=TOXnwoRS6AY

INSTRUCTIONS

- This final assessment contributes 50% to your final grade.
- This assignment is primarily for Course Learning Outcome 2 - Implement a database design group project using appropriate tools such as Oracle SQL.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

Academic Honesty Acknowledgement

"We Sian Yi Xuan**,** Teoh En Yi**,** Kimberly Chan Mun Nga**,** Wong Ke Jie**,** and Fong Khai Shiuan**,** verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. We realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

YX (11 July 2025), EY (2 August 2025), Kim (11 July 2025), KJ (1 August 2025), KS (26 July 2025) (Student signatures/Date)

PART 1: GENERATE A CASE SCENARIO AND DESIGN A RELATIONAL DATA MODEL (35 MARKS)

Scenario

FurEver Home has been recently established as a nonprofit organization for animal welfare services based in Petaling District, Selangor. We provide shelter for many types of animals such as dogs, cats, rabbits, birds, reptiles and small mammals that are stray, abandoned or surrendered. As we perform rescue, rehabilitation and rehoming for these animals, the shelter also aims to raise awareness on pet ownership and good welfare standards.

Due to the complexity of the shelter's daily operations, we have come up with a new mission which is to set up a centralized database system. The main objectives of this database system are to improve overall operational efficiency and ensure ethical standards. By integrating this system, it allows FurEver Home to manage various operations of the shelter effectively. The operations include animal registration, residence assignments, medical records, task assignment, adoptions, donations and staff payroll. We strive for consistency, transparency and accountability throughout the management processes with the combination of multiple workflows and stakeholders.

Our process starts with the animal's arrival where they are immediately registered into the Animal database. Each complete entry includes information like unique ID, name, species, breed, age, gender, arrival date and status. Moving on, the animal will be evaluated for their health condition by our veterinary personnel within three days of their arrival at the animal shelter. The health assessment report is then recorded in the HealthCheck table. The data entry may compromise check date, veterinarian name, diagnosis, treatment plans and numerical health ratings so that this information can be a guideline for subsequent care.

When the animal is cleared from the health assessment process, they will be allocated to suitable living areas which are called Residences. We monitor the cleanliness status and availability of each residence regularly to make sure that hygiene standards in our shelter

are always maintained. The Residences table is also used to track current occupancy levels of each residence so that the capacities are not exceeded to prevent overcrowding and provide healthy living environments for the animals. We also make thorough decisions for the animal placement based on their species' compatibility to minimize stress.

The Task table is used to organize the daily care routines for the animals such as feeding schedules, grooming sessions, exercise programs and medical treatments. We assigned each task to exactly one animal and the task is carried out by one trained volunteer only. The Task table also maintains detailed records of the task assignments which include task ID, volunteer ID, animal ID and task type. This approach can help to ensure consistency in animal care and allow our staff to identify patterns in animal behavior or health. Moreover, volunteers are assigned to tasks based on their availability patterns such as morning, afternoon or evening shifts and the specific needs of animals under their care.

As for the adoption process, eligible animals will first receive vaccinations. This way, we can guarantee safety and adherence to regulations. The data for this process is entered to the Vaccinations table that tracks the vaccination ID, animal ID, vaccination type and vaccination date. Subsequently, prospective adopters can complete their detailed applications through the Adopter table. This table maintains records of adopter personal information such as phone number, email and address. When the application is approved, adoption events are documented in the Adoption table along with the date of adoption and related charges. To encourage responsible pet ownership, an adopter is limited to adopting only a maximum of three animals within a calendar year.

As FurEver Home is a non-profit organization, it heavily depends on public donations and sponsors to meet both general expenses and costs associated with specific animals. Information related to donors is organized in the Donor table, whereas the Donation table logs transaction amounts and dates. As for funds aimed at individual animals, it is monitored through the AnimalDonation junction table. All monetary transactions which include payments for staff salaries are recorded in the FinanceTransaction table and linked to employee compensation records in the PaySlip table.

Additionally, all keepers and veterinary assistant detailed profiles are listed on the Keepers table. This table keeps information about each keeper's role, shift schedule and contact information which are essential for proper staffing coverage. These staff are also assigned to specific animals via the AnimalKeeper junction table. This approach makes sure that all animals receive consistent and quality care from familiar staff members.

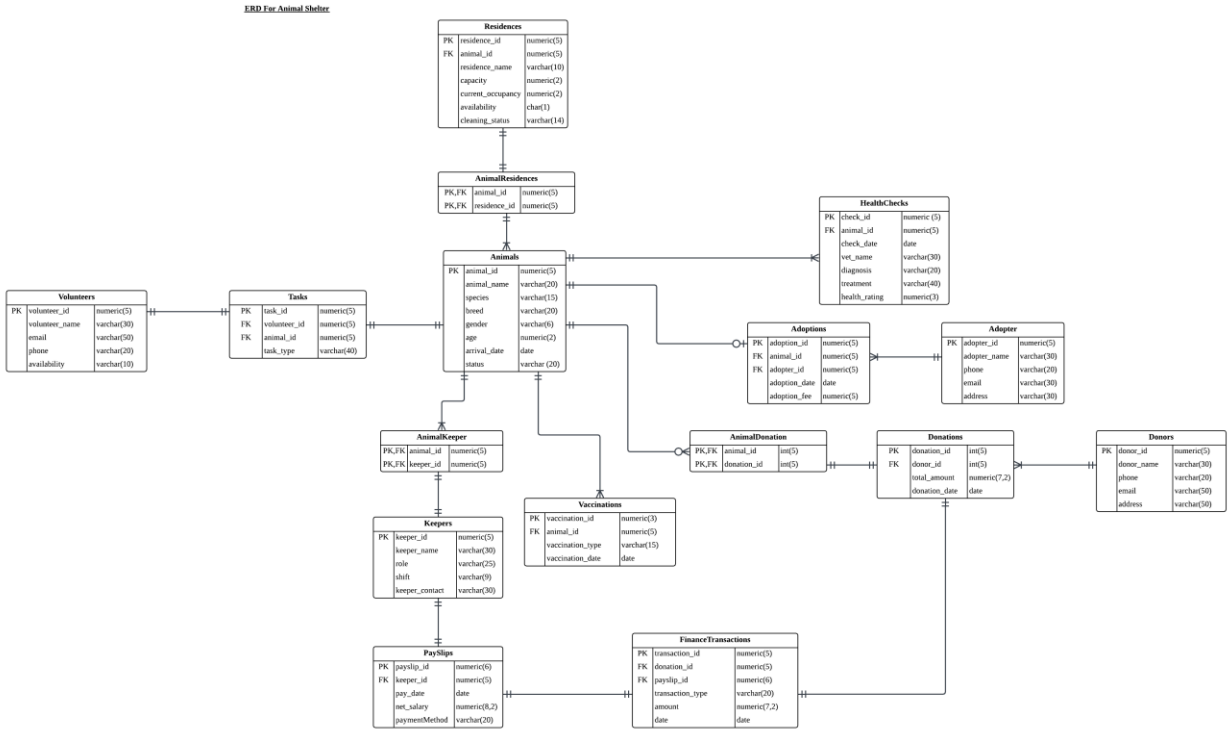
Overall, this database serves as a digital framework that oversees the shelter's daily activities. FurEver Home combines its operational, medical, financial and volunteer workflows into one centralized system. We believe this integration creates a methodical, ethical and data-driven approach to care for animals and ensure their well-being is managed efficiently.

Business Rules

1. Each task is assigned to exactly one animal, and each animal can be associated with only one task.
2. Each task is assigned to exactly one volunteer, and each volunteer can be associated with only one task.
3. A residence can be associated with at least one animal, and each animal can only link to one and only one residence. A junction table named AnimalResidences was created to record the animals staying in a residence.
4. Each animal must have at least one health check, and each health check must be assigned to only one animal. The range of health_rating is from 1 to 5. 1 means the animal is in a very bad state of health, and 5 means the animal is healthy and free from any disease.
5. Each animal can only have one adoption or no adoption, and each adoption can only have one animal.
6. An adopter can have at least one adoption, and each adoption can only have one adopter.

7. Each animal can be associated with zero or many donations, each donation must be associated with one and only one animal. A junction table named AnimalDonation was created to record which donation goes to which animal.
8. A donor must have at least one donation, and each donation comes from a donor.
9. Each animal must have at least one vaccination, and each vaccination related to exactly an animal.
10. Each keeper can be associated with multiple animals, and each animal can be associated with multiple keepers. The AnimalKeeper junction table tracks these many-to-many relationships to ensure proper care coordination and accountability.
11. Each keeper is associated with exactly one pay slip. Each pay slip is also associated with exactly one keeper.
12. Each finance transaction is associated with one and only one pay slip or donation, and each pay slip or donation is associated with exactly one finance transaction. Donation_id and payslip_id are nullable. If the transaction_type is donation, payslip_id will be null. If the transaction_type is payslip, donation_id will be null.

Entity Relationship Diagram



Constraints

	Constraint	Justification
1	Health assessment must be done within 3 days of animal arrival	It encourages quick medical examination of all incoming animals to stop the spread of disease and satisfy legal obligations for animal shelters. By recognizing critical cases and upholding organized health monitoring procedures, this limitation maximizes the use of available resources.
2	Animal must be vaccinated before adoption	Ensures that adoption laws are followed legally and only animals with up-to-date vaccinations are allowed to be adopted to safeguard the public's health. As it eliminates liability concerns, lowers the danger of disease transmission and exemplifies proper shelter administration, this restriction can raise adoption rates.
3	An adopter can adopt up to 3 animals per calendar year	Reduces the possibility of neglect or inadequate care while upholding responsible ownership boundaries. This restriction also helps optimize resource allocation, guarantees sufficient care capacity and prevents hoarding situations.
4	Volunteers are assigned tasks based on their availability patterns (morning, afternoon, evening)	Ensures operational efficiency, maintains adequate staffing levels based on volunteer availability and maximizes volunteer satisfaction through appropriate task matching
5	Each finance transaction must be linked to either a	Helps avoid accounting errors and guarantees the integrity of financial data. By classifying revenue and expenses, this constraint enables accurate financial reporting, avoids transaction double counting and preserves transparent

	donation or a payslip. It can never be both.	audit trails. It also increases database performance by eliminating redundant relationships and ensures regulatory compliance for nonprofit financial reporting.
6	Shelter occupancy cannot exceed its capacity	Maintains the operational limits for physical shelter space and the safety of the animals under our care. This constraint prevents overcrowding that may result in disease outbreaks, behavioral issues, and safety hazards. It makes sure that all animals receive proper care while preserving operational efficiency and optimizes resource allocation.
7	Adoption Fee must be non-negative	Prevents invalid financial values and maintains the integrity of financial data. This constraint preserves proper financial reporting, helps with audit compliance and avoids revenue loss due to negative value entries.
8	Animal age must be positive and cannot exceed 100 years	Allows for realistic age data entry while taking into account long-lived species such as parrots and tortoises. It supports accurate demographic reporting and eliminates values that are unrealistic.
9	Vaccination records must be maintained for at least 7 years after animal departure	Fulfills legal record-keeping or auditing requirements. This constraint satisfies regulatory record-keeping mandates, supports liability protection and enables historical health tracking. It optimizes data retention policies while maintaining comprehensive health records for legal and operational purposes.
10	Donation amounts must be greater than RM0	Ensures financial data integrity and prevents accounting errors. This constraint prevents invalid donation entries that could corrupt financial reports, maintain accurate donor records and support proper tax reporting.

PART 2: IMPLEMENT AND QUERY A DATABASE USING SQL (30 MARKS)

a) Outer Join Query

SQL:

```sql
SELECT
    a.animal_id,
    a.animal_name,
    a.species,
    a.breed,
    a.status,
    ad.adoption_date,
    adopter.adopter_name,
    ad.adoption_fee
FROM Animals a
LEFT OUTER JOIN Adoption ad ON a.animal_id = ad.animal_id
LEFT OUTER JOIN Adopter adopter ON ad.adopter_id = adopter.adopter_id
ORDER BY a.animal_id;
```

Output:

ANIMAL_ID	ANIMAL_NAME	SPECIES	BREED	STATUS	ADOPTION_DATE	ADOPTER_NAME	ADOPTION_FEE
101	Buddy	Dog	Golden Retriever	Adopted	15-JAN-2023	Robert Adward	150
102	Mittens	Cat	Siamese	Adopted	20-FEB-2023	Jennifer Smith	150
103	Rocky	Parrot	African Grey	Adopted	25-FEB-2023	Bella Hadid	200
104	Luna	Rabbit	Holland Lop	Available	-	-	-
105	Charlie	Guinea Pig	American	Adopted	28-FEB-2023	Lisa Brown	100
106	Bella	Dog	Labrador	Adopted	05-MAR-2023	Christopher Davis	100
107	Simba	Lizard	Bearded Dragon	Adopted	10-MAR-2023	Jenny Jane	75
108	Daisy	Hamster	Syrian	Medical Hold	-	-	-
109	Max	Dog	German Shepherd	Adopted	15-MAR-2023	Matthew Wilson	100
110	Whiskers	Cat	Persian	Available	-	-	-
111	Polly	Parrot	Macaw	Available	-	-	-
112	Oscar	Bird	Cockatiel	Available	-	-	-
113	Peanut	Guinea Pig	Abyssinian	Adopted	13-APR-2023	Jeff Satur	100
114	Cooper	Dog	Beagle	Available	-	-	-
115	Cleo	Cat	Maine Coon	Available	13-APR-2023	Jeff Satur	300

15 rows returned in 0.01 seconds Download

Justification:

For the animal shelter system to track adoptions and manage an extensive animal inventory, this query is necessary. Only the 9 adopted animals would be displayed by an INNER JOIN, excluding the six unadopted animals. The LEFT OUTER JOIN query shows every animal regardless of adoption status in the results. Because of this visibility, management can determine adoption success rates, find the animals that have been on the waiting list the longest and efficiently use marketing funds to raise awareness of unadopted animals. When animals have not been adopted, the outer join structure assigns NULL values to adoption-related fields, providing clear visual markers for workers to distinguish between adopted and available animals rapidly. This comprehensive view supports strategic decision-making for animal placement programs, helps identify trends in adoption preferences by species and ensures no animal is overlooked in care planning or promotional activities.

b) 4-Table Join with GROUP BY

SQL:

```sql
SELECT
    a.species,
    COUNT(DISTINCT d.donor_id) as total_donors,
    COUNT(ad.donation_id) as number_of_donations,
    SUM(ad.support_amount) as total_support_received,
    AVG(ad.support_amount) as average_donation_per_animal
FROM Animals a
INNER JOIN AnimalDonation ad ON a.animal_id = ad.animal_id
INNER JOIN Donation d ON ad.donation_id = d.donation_id
INNER JOIN Donor donor ON d.donor_id = donor.donor_id
GROUP BY a.species
HAVING SUM(ad.support_amount) > 200
ORDER BY total_support_received DESC;
```

Output:

SPECIES	TOTAL_DONORS	NUMBER_OF_DONATIONS	TOTAL_SUPPORT_RECEIVED	AVERAGE_DONATION_PER_ANIMAL
Dog	4	4	1550	387.5
Bird	2	2	1400	700
Parrot	2	2	1100	550
Rabbit	1	1	750	750
Lizard	1	1	600	600
Cat	2	2	500	250
Hamster	1	1	350	350
Guinea Pig	1	1	300	300

8 rows returned in 0.07 seconds Download

Justification:

The purpose of this SQL query is to analyze donation patterns by animal species across multiple donors including donor information. This multi-table analysis helps with strategic fundraising and resource allocation decisions within the shelter. By joining Animals, AnimalDonation, Donation, and Donor tables with GROUP BY functionality, management can identify which animal species generate the most donor engagement and financial support. The query reveals critical insights about whether dogs receive more donations than cats or if exotic animals like parrots and reptiles require additional marketing efforts to attract financial support. The aggregation functions provide total donors per species, donation frequency, total support amounts and average donation values which are essential for budget planning and species-specific care programs. The HAVING clause filtering for species with over $200 in support helps focus attention on viable funding sources while identifying underperforming categories that may need intervention. This analysis directly supports grant applications by demonstrating community support levels, guides marketing budget allocation across different animal programs, and helps establish sustainable funding models for species-specific veterinary care.

c) String Pattern Matching & Date Functions

SQL:

```sql
SELECT
    v.volunteer_name,
    v.email,
    v.availability,
    t.task_type,
    a.animal_name,
    ROUND(SYSDATE - a.arrival_date) AS days_since_arrival
FROM Volunteers v
INNER JOIN Tasks t ON v.volunteer_id = t.volunteer_id
INNER JOIN Animals a ON t.animal_id = a.animal_id
WHERE v.email LIKE '%gmail.com'
  AND a.arrival_date >= DATE '2023-01-01'
  AND a.arrival_date < DATE '2023-02-01'
ORDER BY v.volunteer_name;
```

Output:

VOLUNTEER_NAME	EMAIL	AVAILABILITY	TASK_TYPE	ANIMAL_NAME	DAYS_SINCE_ARRIVAL
Ava White	ava.w@gmail.com	Afternoon	Exercise	Whiskers	914
Daniel Kim	daniel.k@gmail.com	Monrning	Feeding	Bella	927
David Wilson	david.w@gmail.com	Evening	Exercise	Luna	931
Emily Davis	emily.d@gmail.com	Afternoon	Medical Check	Rocky	929
Ethan Moore	ethan.m@gmail.com	Afternoon	Medical Check	Peanut	916
Grace Hall	grace.h@gmail.com	Morning	Feeding	Oscar	910
James Taylor	james.t@gmail.com	Evening	Medical Check	Daisy	923
Jessica Lee	jessica.l@gmail.com	Afternoon	Grooming	Charlie	925
Liam Harris	liam.h@gmail.com	Evening	Grooming	Polly	918
Lucas Scott	lucas.s@gmail.com	Morning	Play Time	Cleo	907
Michael Brown	michael.b@gmail.com	Morning	Bathing	Mittens	933
Nathan Green	nathan.g@gmail.com	Morning	Cleaning	Max	921
Olivia Martinez	olivia.m@gmail.com	Afternoon	Training	Simba	919
Sarah Nelson	sarah.nel@gmail.com	Morning	Feeding	Buddy	935
Sophie King	sophie.k@gmail.com	Evening	Training	Cooper	912

Justification:

The purpose of this SQL query is to find all volunteers with Gmail accounts who are scheduled for tasks for animals that arrived in January 2023 and show how many days since their task assignment .This query serves multiple operational purposes for the animal shelter by identifying volunteers with Gmail accounts using pattern matching (LIKE '%gmail.com'). The shelter also can implement targeted communication strategies with specific volunteer groups and tracks task assignment timelines. The date functions calculate how long each animal has been at the shelter

(SYSDATE - arrival_date) which is crucial for prioritizing care and adoption efforts. Animals with longer stays may require additional medical attention, behavioral training or enhanced marketing efforts to find suitable homes. This combination allows management to efficiently coordinate volunteer assignments with animal care priorities, ensuring that experienced volunteers can be matched with animals requiring more attention due to extended shelter stays. Additionally, this data supports performance evaluation by tracking volunteers who consistently work with challenging long-term residents versus newly arrived animals.

d) A query having both OR and AND

SQL:

```
SELECT a.animal_id, a.animal_name, a.species, a.status, hc.health_rating
FROM Animals a
JOIN HealthCheck hc ON a.animal_id = hc.animal_id
WHERE (a.species = 'Dog' OR a.species = 'Cat')
  AND hc.health_rating >= 4
  AND a.status = 'Available'
  ORDER BY hc.health_rating desc;
```

Output:

ANIMAL_ID	ANIMAL_NAME	SPECIES	STATUS	HEALTH_RATING
114	Cooper	Dog	Available	5
115	Cleo	Cat	Available	4
110	Whiskers	Cat	Available	4

Justification:

The purpose of this SQL query is to find out dogs or cats in the shelter that are available for adoption and have a positive health rating which is 4 and above. It uses OR and AND conditions to filter animal species and health ratings, together with adoption status. The query result helps in identifying commonly adopted animals like dogs or cats that are healthy and available for adoption, sorted by the most positive health rating. This query is important in prioritizing animals that have the highest chances of being adopted. The shelter could also utilize the query result to promote adoption events for these animals.

e) A query that consists of at least 2 subqueries

SQL:

```
SELECT k.keep_id as keeper_id, k.keeper_name, k.role, ps.net_salary AS salary, a.animal_name, a.species, a.status, hc.health_rating
FROM
    (SELECT *
     FROM Keepers
     WHERE role = 'Medical Assistant') k
JOIN PaySlip ps ON k.keep_id = ps.keeper_id
JOIN AnimalKeeper ak ON k.keep_id = ak.keeper_id
JOIN Animals a ON ak.animal_id = a.animal_id
JOIN (
    SELECT hc1.*
    FROM HealthCheck hc1
    JOIN (
        SELECT animal_id, MAX(check_date) AS latest_date
        FROM HealthCheck
        GROUP BY animal_id
    ) latest ON hc1.animal_id = latest.animal_id AND hc1.check_date = latest.latest_date
) hc ON a.animal_id = hc.animal_id
ORDER BY ps.net_salary DESC;
```

Output:

KEEPER_ID	KEEPER_NAME	ROLE	SALARY	ANIMAL_NAME	SPECIES	STATUS	HEALTH_RATING
1004	Susan Hernandez	Medical Assistant	3500	Luna	Rabbit	Available	5
1019	Anthony Perez	Medical Assistant	3300	Daisy	Hamster	Medical Hold	2
1001	John Garcia	Medical Assistant	3200	Buddy	Dog	Adopted	5
1015	Emma Walker	Medical Assistant	3000	Cleo	Cat	Available	4
1011	Patricia Kim	Medical Assistant	2700	Polly	Parrot	Available	5

Justification:

The purpose of this SQL query is to analyze and evaluate the performance and compensation of keepers who have the role of Medical Assistant. It combines the data of their salary with the status and latest health condition of the animals they take care of. The first subquery is used to filter the keepers who are Medical Assistant, and the second one to extract the status and health rating of their assigned animals. The result of this query helps the shelter to evaluate whether a medical assistant's compensation is appropriate based on the health status of their assigned animals. This query is useful for performance review and resource planning for the shelter.

f) List the animals that have been in the sanctuary for over a year and have received the most positive health check ratings for health studies

SQL:

```
SELECT a.animal_id, a.animal_name, a.species, a.arrival_date, hc.health_rating
FROM Animals a
JOIN HealthCheck hc ON a.animal_id = hc.animal_id
WHERE a.status <> 'Adopted'
  AND a.arrival_date <= ADD_MONTHS(SYSDATE, -12)
  AND hc.health_rating = 5
ORDER BY a.arrival_date;
```

Output:

ANIMAL_ID	ANIMAL_NAME	SPECIES	ARRIVAL_DATE	HEALTH_RATING
104	Luna	Rabbit	05-JAN-2023	5
111	Polly	Parrot	18-JAN-2023	5
114	Cooper	Dog	24-JAN-2023	5
112	Oscar	Bird	26-JAN-2023	5

Justification:

The purpose of this SQL query is to identify animals that have been in the shelter for more than one year but not adopted, and have the most positive health rating which is 5 in health checkup. This query is helpful in determining animals that are in healthy condition and still available for adoption. The query result is useful in prioritizing animals that stayed in the shelter for a long time to receive promotions and showcase them in adoption campaigns.

PART 4: DEMONSTRATION (10 MARKS)

List each student's presentation here. Please update the following table accordingly.

No	ID	Name	Questions to present
i)	24120016	Wong Ke Jie	Project's goal and database solution
ii)	23099609	Sian Yi Xuan	Explain about the design of our database
iii)	23051105	Teoh En Yi	Running and explain about the SQL script
iv)	23098643	Kimberly Chan	Explain and run SQL queries a,b,c
v)	23074925	Fong Khai Shiuan	Explain and run SQL queries d,e,f